

# Project 1

## Advection Equation and Solar Wind Problem

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### NUMERICAL MODELING OF THE PARKER SOLAR WIND USING THE ADVECTION EQUATION

This project introduces students to numerical methods for solving ordinary differential equations (ODEs) in the context of astrophysical fluid dynamics. You will focus on the Parker solar wind model, where the advection term in the momentum equation plays a key role in describing the acceleration of plasma from the Sun. The project builds on the isothermal, spherically symmetric hydrodynamic model, incorporating advection to achieve transonic outflow. You will implement a numerical solver, perform parameter studies, and validate results through plots and comparisons. Use Python with libraries such as NumPy and Matplotlib for visualization (avoid pre-built ODE solvers like SciPy's `solve_ivp` or `odeint`; instead, implement manual numerical integration methods as described below). The project should not require advanced MHD simulations but should involve thoughtful code design and analysis.

#### BACKGROUND READING

Review the step-by-step derivation provided in the lecture notes (or equivalent resources like Parker's original 1958 paper or modern reviews). Key equations include:

- Momentum:

$$v \frac{dv}{dr} + \frac{1}{\rho} \frac{dP}{dr} + \frac{GM_{\odot}}{r^2} = 0 \quad \text{advection term: } v \frac{dv}{dr} \quad (0.1)$$

- Continuity:

$$4\pi r^2 \rho v = \dot{M} \quad (\text{constant mass flux}) \quad (0.2)$$

- Isothermal:

$$P = \frac{RT}{\mu} \rho \quad \text{leading to the integrated transonic solution.} \quad (0.3)$$

#### NUMERICAL METHODS GUIDANCE

Instead of using built-in solvers like `solve_ivp` or `odeint`, implement manual numerical integration for the ODE. The core ODE to solve is:

$$\frac{dv}{dr} = \frac{\frac{2v_c^2}{r} - \frac{GM_\odot}{r^2}}{v - \frac{v_c^2}{v}} \quad (0.4)$$

where  $v_c = \sqrt{\frac{k_B T}{\mu m_p}}$  (using Boltzmann constant  $k_B$ , proton mass  $m_p$ ), or equivalently with gas constant  $\mathcal{R}$ . To handle the singularity at the critical point ( $r_c$ , where  $v = v_c$ ), integrate the subsonic branch backward from just below  $r_c$  to  $r_{\min}$ , and the supersonic branch forward from just above  $r_c$  to  $r_{\max}$ , then combine.

USE THE FOLLOWING MANUAL METHODS

**Euler Method (Riemann Sum Approximation):** This is a basic first-order method that approximates the integral  $\Delta v \approx f(r, v) \Delta r$  using left Riemann sums for forward steps (or right for backward). It's simple but may require small step sizes for accuracy near the critical point.

**Runge-Kutta 4th Order (RK4):** A higher-order method that improves accuracy by weighted averages, akin to Simpson's rule in quadrature (which uses parabolic approximations instead of rectangles in Riemann sums). RK4 evaluates the function at multiple points per step, providing better stability for stiff ODEs like this one. For both methods, use small perturbations (epsilon  $-10^{-3} r_c$  for  $r$ ,  $10^{-3} v_c$  for  $v$ ) at the critical point to start integration. Convert all units to SI (m, m/s) for computation, then plot in  $R_\odot$  and km/s.

THEORETICAL SETUP (ANALYTICAL WARM-UP)

DERIVE THE CRITICAL RADIUS AND SONIC SPEED

Starting from the momentum and continuity equations, derive the expression for the critical radius

$$r_c = \frac{GM_\odot}{2v_c^2} \quad (0.5)$$

where

$$v_c = \sqrt{\frac{k_B T}{\mu m_p}} \quad \text{or equiv. with } \mathcal{R} \quad (0.6)$$

Assume solar parameters:

$$\begin{aligned} M_\odot &= 1.989 \times 10^{30} \text{ kg} \\ G &= 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \\ k_B &= 1.380649 \times 10^{-23} \text{ J K}^{-1} \\ m_p &= 1.6726219 \times 10^{-27} \text{ kg} \\ \mu &= 0.5 \quad (\text{for fully ionized hydrogen}) \\ T &= 10^6 \text{ K} \\ R_\odot &= 6.96 \times 10^8 \text{ m.} \end{aligned}$$

Calculate numerical values for  $v_c$  (128 km/s) and  $r_c$  (5.8  $R_\odot$ ). Discuss why the advection term is essential for the wind to escape gravity.

ANALYTICAL SOLUTION PLOT

Using the integrated form:

$$\left(\frac{v}{v_c}\right)^2 - \ln\left(\frac{v}{v_c}\right) = 4 \ln\left(\frac{r}{r_c}\right) + 4 \frac{r_c}{r} - 3 \quad (0.7)$$

Solve for  $v(r)$  semi-analytically using root-finding (e.g., Newton-Raphson or Bisection methods) over  $r$  from  $1R_\odot$  to  $100R_\odot$ . Plot  $v(r)$  in km/s vs.  $r$  in  $R_\odot$ . Mark the critical point and label subsonic/supersonic regions. Compare the asymptotic velocity at large  $r$  to observed solar wind speeds ( $\sim 400$  km/s for slow wind).

## NUMERICAL IMPLEMENTATION OF THE ODE SYSTEM

### FORMULATE THE ODE SYSTEM

Rewrite the governing equations as a first-order ODE as above. Include density

$$\rho(r) = \frac{\dot{M}}{4\pi r^2 v(r)} \quad (0.8)$$

Choose  $\dot{M} = 2 \times 10^{-14} M_{\odot} \text{ yr}^{-1}$  (convert to kg/s:  $\sim 1.26 \times 10^9 \text{ kg/s}$ ). Explain how to handle the sonic point singularity using separate integrations and small perturbations.

### ODE DEVELOPMENT: BASELINE SOLVER

Implement both Euler and RK4 using the pseudocode above in a function 'solve\_parker\_wind(T, r\_min=1.5, r\_max=100, h=1e6, method='rk4')' that takes temperature  $T$ , returns arrays for  $r$  ( $R_{\odot}$ ),  $v(r)$  (km/s),  $\rho(r)$  (kg/m<sup>3</sup>). Test with  $T = 10^6$  K; plot  $v(r)$  for both methods overlaid on your analytical solution from Part 1. Compute relative error at  $r = 50R_{\odot}$ . Discuss issues (e.g., Euler instability with large  $h$ , RK4 better handling stiffness) and mitigation (e.g., adaptive  $h$  near  $r_c$ ).

## PARAMETER STUDIES AND VISUALIZATION

### TEMPERATURE SENSITIVITY ANALYSIS

Using RK4, run for  $T = [0.5 \times 10^6, 1 \times 10^6, 2 \times 10^6] \text{ K}$ . Plot  $v(r)$  (multi-line), mark  $r_c$ , and  $\rho(r)$  (log-log). Explain  $T$ 's effect on advection and wind at 1 AU ( $215 R_{\odot}$ ). Find min  $T$  for transonic wind numerically (e.g., where solutions don't accelerate).

### MASS LOSS RATE VARIATION

Fix  $T = 10^6 \text{ K}$ , vary  $\dot{M}$  by 0.5x and 2x. Plot  $v(r)$ ,  $\rho(r)$ . Verify  $v$  independent of  $\dot{M}$ ,  $\rho$  linear; discuss advection's role. Table: parameters at 1 AU.

## EXTENSION AND VALIDATION

### SIMPLE EXTENSION: INCLUDE A BASIC HEATING TERM

Add  $+\frac{Q(r)}{v}$  to  $dv/dr$  numerator ( $Q(r) = Q_0 \exp(-r/H)$ ,  $H = 5R_{\odot}$ , tune  $Q_0$  for +20% terminal  $v$ ). Reimplement with RK4; compare plots. Discuss advection-energy interplay.

**Final Remarks:** Encourage collaboration on concepts but individual code/report. This project reinforces computational skills while connecting to real astrophysical phenomena!

## SUBMISSION REQUIREMENTS

Submit Jupyter notebooks or Python scripts with code, plots, and report discussing results, numerical challenges (e.g., stability near the critical point, comparison of Euler vs. RK4 accuracy/convergence), and physical interpretations. Include comments in your code for clarity. Compare Euler and RK4 results in your report (e.g., relative errors vs. analytical solution).